3.2 Mechanical properties of materials

This section is about materials and how their mechanical properties (and hence their applications) are related to their structures. The physics may be put into context through a study of materials in medicine and engineering. Human and cultural issues arise in considering the impact of materials on technology and society (HSW7, 9, 10, 11, 12).

It is not intended that learners acquire a detailed knowledge of a range of materials. Learners should have a reading comprehension of terms needed to understand accounts of structure, uses and properties of materials. Examples should include: a metal, a ceramic and a long-chain polymer. Learners should be given opportunities to discuss and share their understanding of the uses and properties of these materials (HSW8).

		Learning outcomes	Additional guidance
(a)		*Describe and explain:*	
	(i)	simple mechanical behaviour: elastic and plastic deformation and fracture	
	(ii)	direct evidence of the size of particles and their spacing	Examples: Scanning Tunnelling Microscope images; Rayleigh's oil drop experiment HSW7, 11
	(iii)	behaviour/structure of classes of materials, limited to metals, ceramics and polymers; dislocations leading to slip in metals with brittle materials not having mobile dislocations; polymer behaviour in terms of chain entanglement/unravelling	HSW12
	(iv)	one method of measuring Young modulus and fracture stress.	*M1.2*
(b)		*Make appropriate use of:*	
	(i)	the terms: stress, strain, Young modulus, tension, compression, fracture stress and yield stress, stiff, elastic, plastic, ductile, hard, brittle, tough, strong, dislocation	HSW8
		by sketching and interpreting:	
	(ii)	force–extension and stress–strain graphs up to fracture	*M3.1, M3.2, M3.12*
	(iii)	tables and diagrams comparing materials by properties	*M3.1, M3.10*
	(iv)	images showing structures of materials.	HSW7, 11

(c) *Make calculations and estimates involving:*

(i) Hooke's Law, $F = kx$; energy stored in an elastic material (elastic strain energy) $= \frac{1}{2}kx^2$; energy as area under Force–extension graph for elastic materials	*M0.5, M2.2, M2.3, M3.8* (loading only)
(ii) stress $= \dfrac{\text{tension}}{\text{cross-sectional area}}$, strain $= \dfrac{\text{extension}}{\text{original length}}$, Young modulus $E = \dfrac{\text{stress}}{\text{strain}}$	*M0.3, M1.1, M1.4*

(d) *Demonstrate and apply knowledge and understanding of the following practical activities:*

(i) plotting force–extension characteristics for arrangements of springs, rubber bands, polythene strips, etc.	links to **3.2a(iii)**, **b(ii)**, **c(i)** HSW4 **PAG2**
(ii) determining Young modulus for a metal such as copper, steel or other wire.	links to **3.2a(iv)**, **c(ii)** **PAG2**